

Name:

BUID:

Lab 9: Progress and Preservation

1. Recall the simply-typed lambda calculus (STLC) with small-step CBN semantics.

$e ::= \bullet \mid \lambda x^\tau. e \mid ee$ (expressions)

$\tau ::= \top \mid \tau \rightarrow \tau$ (types)

$v ::= \bullet \mid \lambda x^\tau. e$ (values)

$$\begin{array}{c} \frac{}{\Gamma \vdash \bullet : \top} \text{UNIT} \quad \frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau} \text{VAR} \quad \frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \lambda x^{\tau_1}. e : \tau_1 \rightarrow \tau_2} \text{FUN} \quad \frac{\Gamma \vdash e_1 : \tau_2 \rightarrow \tau \quad \Gamma \vdash e_2 : \tau_2}{\Gamma \vdash e_1 e_2 : \tau} \text{APP} \\ \\ \frac{e_1 \longrightarrow e'_1}{e_1 e_2 \longrightarrow e'_1 e_2} \text{APPLEFT} \quad \frac{}{(\lambda x^\tau. e_1) e_2 \longrightarrow [e_2/x] e_1} \text{BETA} \end{array}$$

Suppose we made a mistake when writing the FUN rule, and forgot to require the type annotation to match the argument type:

$$\frac{\Gamma, x : \tau \vdash e : \tau_2}{\Gamma \vdash \lambda x^\tau. e : \tau_1 \rightarrow \tau_2} \text{FUNWRONG}$$

Demonstrate that STLC with FUN replaced by FUNWRONG does not satisfy preservation.

2. Suppose we consider a variation of STLC that requires applications to have two arguments.

$e ::= \bullet \mid \lambda x^\tau. e \mid e(e, e)$ (expressions)

$\tau ::= \top \mid \tau \rightarrow \tau$ (types)

$v ::= \bullet \mid \lambda x^\tau. e$ (values)

$$\begin{array}{c} \frac{}{\Gamma \vdash \bullet : \top} \text{UNIT} \quad \frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau} \text{VAR} \quad \frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \lambda x^{\tau_1}. e : \tau_1 \rightarrow \tau_2} \text{FUN} \quad \frac{\Gamma \vdash e : \tau_1 \rightarrow \tau_2 \rightarrow \tau \quad \Gamma \vdash e_1 : \tau_1 \quad \Gamma \vdash e_2 : \tau_2}{\Gamma \vdash e(e_1, e_2) : \tau} \text{APP} \\ \\ \frac{e \longrightarrow e'}{e(e_1, e_2) \longrightarrow e'(e_1, e_2)} \text{APPLEFT} \quad \frac{}{(\lambda x^{\tau_1} \lambda y^{\tau_2}. e)(e_1, e_2) \longrightarrow [e_2/y][e_1/x] e} \text{BETA} \end{array}$$

Demonstrate that this variant of STLC does not satisfy progress.

3. *Optional Challenge.* Show by induction on derivations that the system from Problem 1 satisfies progress¹

4. *Optional Challenge.* Show by induction on derivations that the system from Problem 2 satisfies preservation²

Write your solutions below and on the back on this page.

1. $\emptyset \vdash (\lambda x^\top. x)(\lambda x^\top. x) : \top$ $(\lambda x^\top. x)(\lambda x^\top. x) \longrightarrow_\beta \lambda x^\top. x$

$$\begin{array}{l} \vdash \emptyset \vdash \lambda x^\top. x : (\top \rightarrow \top) \rightarrow \top \\ \quad \vdash \{x : \top\} \vdash x : \top \\ \vdash \emptyset \vdash \lambda x^\top. x : \top \rightarrow \top \\ \quad \vdash \{x : \top\} \vdash x : \top \end{array}$$

$$\vdash \lambda x^\top. x : \top \rightarrow \top$$

¹This challenge problem implies progress does not imply preservation.

²This challenge problem implies preservation does not imply progress.

Continue your work here.

$$2. (\lambda x^{T \rightarrow T}. x) (\lambda x^T. x, \bullet)$$

$$(\lambda x^{T \rightarrow T}. x) (\lambda x^T. x, \bullet) \rightarrow_e$$

$$\emptyset \vdash (\lambda x^{T \rightarrow T}. x) (\lambda x^T. x, \bullet) : T \quad (\text{app})$$

$$\vdash \emptyset \vdash \lambda x^{T \rightarrow T}. x : (T \rightarrow T) \rightarrow T \rightarrow T \quad (\text{fun})$$

$$\quad \vdash \{x : T \rightarrow T\} \vdash x : T \rightarrow T \quad (\text{var})$$

$$\vdash \emptyset \vdash \lambda x^T. x : T \rightarrow T \quad (\text{fun})$$

$$\quad \vdash \{x : T\} \vdash x : T \quad (\text{var})$$

$$\vdash \emptyset \vdash \bullet : T \quad (\text{unit})$$